

TELLCO TELECOM

Acquisition Due Diligence

A data-driven view of user overview, engagement, network experience, and satisfaction

Why This Analysis Matters

Our investor specializes in acquiring undervalued assets and driving profitability by refocusing the business on its most valuable customers and products. TellCo's current owners have shared financial data but have never analyzed their own system-generated network data.

- One month of xDR (session-level) network data analyzed
- 106,857 unique subscribers, 150,001 sessions
- Four analysis stages: Overview, Engagement, Experience, Satisfaction
- Goal: identify growth levers and a buy/sell recommendation

Precedent

On a prior engagement, this same investor used rich data analysis to identify that university-student delivery routes were the most profitable segment for a delivery company -- driving a 25% profit increase in 6 months after acquisition. The same rigor is applied here to TellCo.

Conditional Buy

A large, actively engaged, premium-skewed subscriber base with clear monetization levers -- offset by network telemetry gaps and quality issues a buyer should price in.

106,857

Unique subscribers

150,001

xDR sessions analyzed

58.7 TB

Gaming data volume (top app)

95%

Satisfaction variance explained by usage

- Premium device base (Apple/Samsung/Huawei) signals real spending power
- Distinct fixed-wireless broadband segment (Huawei router users) is underexploited
- Network quality varies materially by handset -- remediation opportunity
- Recommend a price adjustment or escrow for network-infrastructure investment

Methodology & Data Notes

1

Clean

Missing values imputed with column mean/mode; extreme outliers capped at 3x IQR

2

Aggregate

Session-level xDR rows rolled up to per-subscriber (MSISDN) features

3

Model

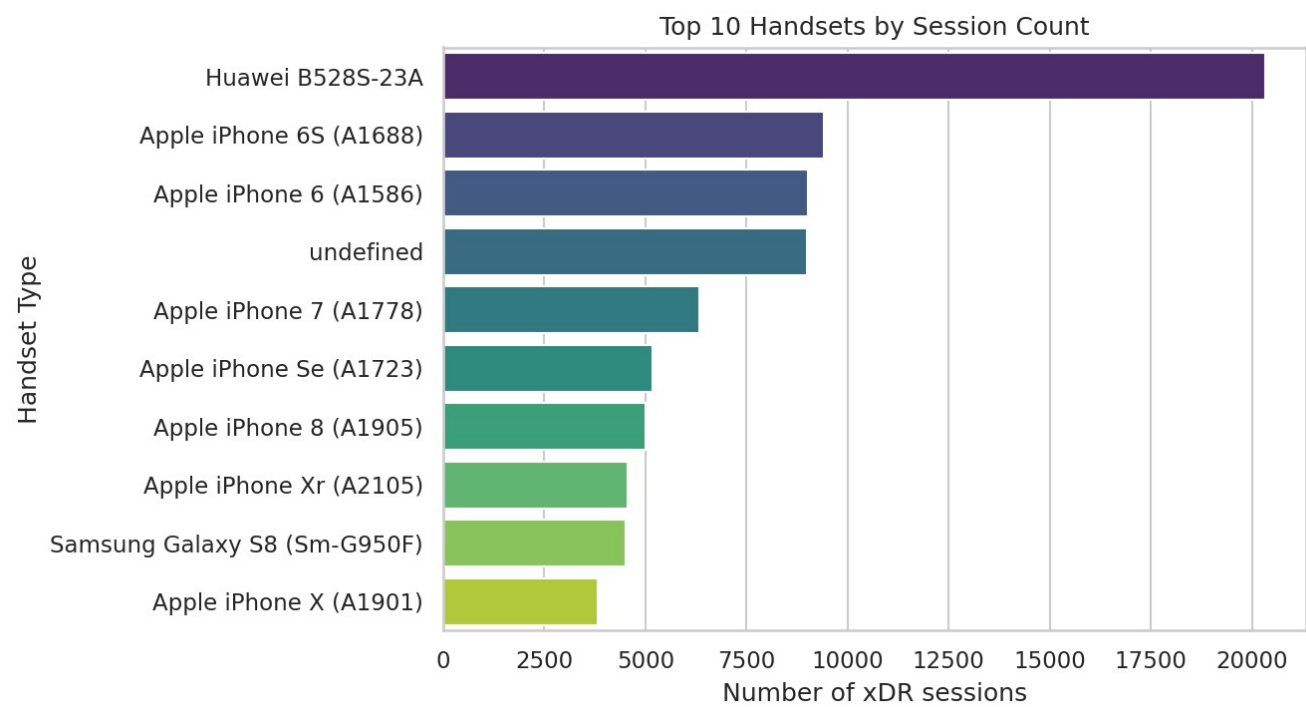
K-Means clustering, PCA, and linear regression applied to engagement & experience metrics

4

Score

Engagement + experience distances combined into a satisfaction score per subscriber

Handset & Manufacturer Landscape



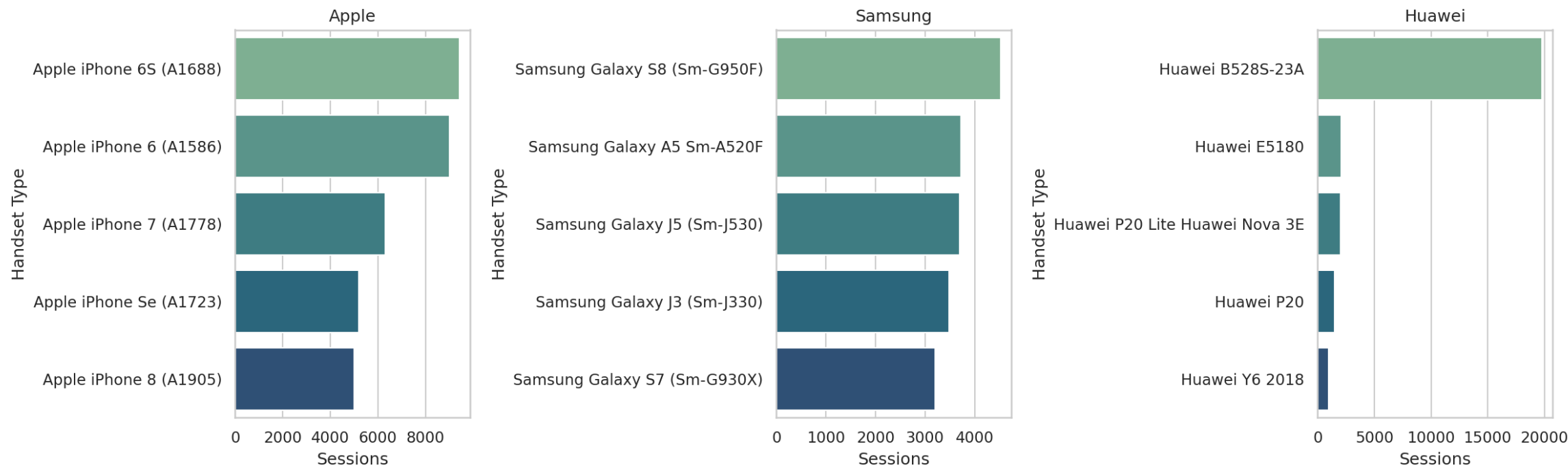
Top 3 Manufacturers

60,137
Apple

40,839
Samsung

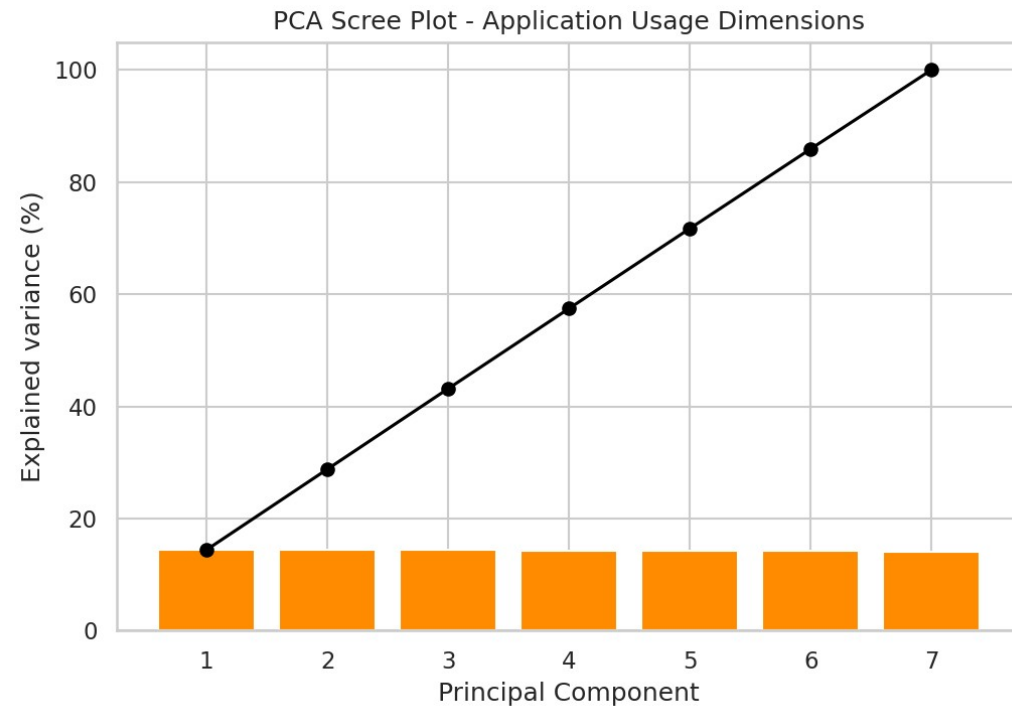
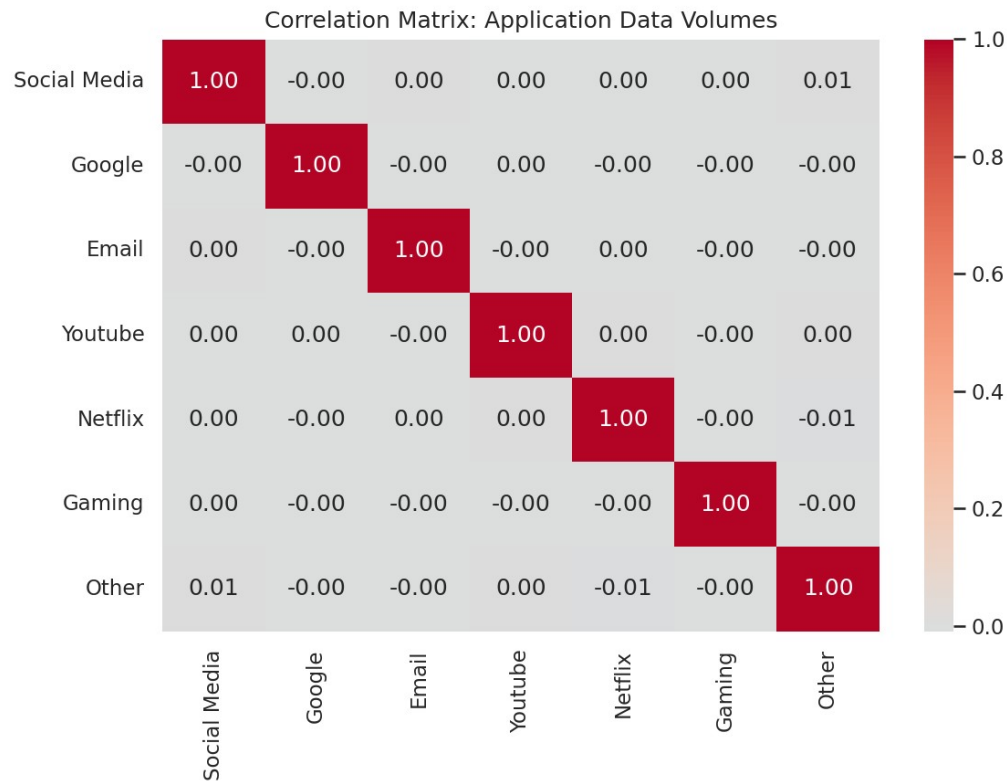
34,423
Huawei

Top 5 Handsets per Top-3 Manufacturer



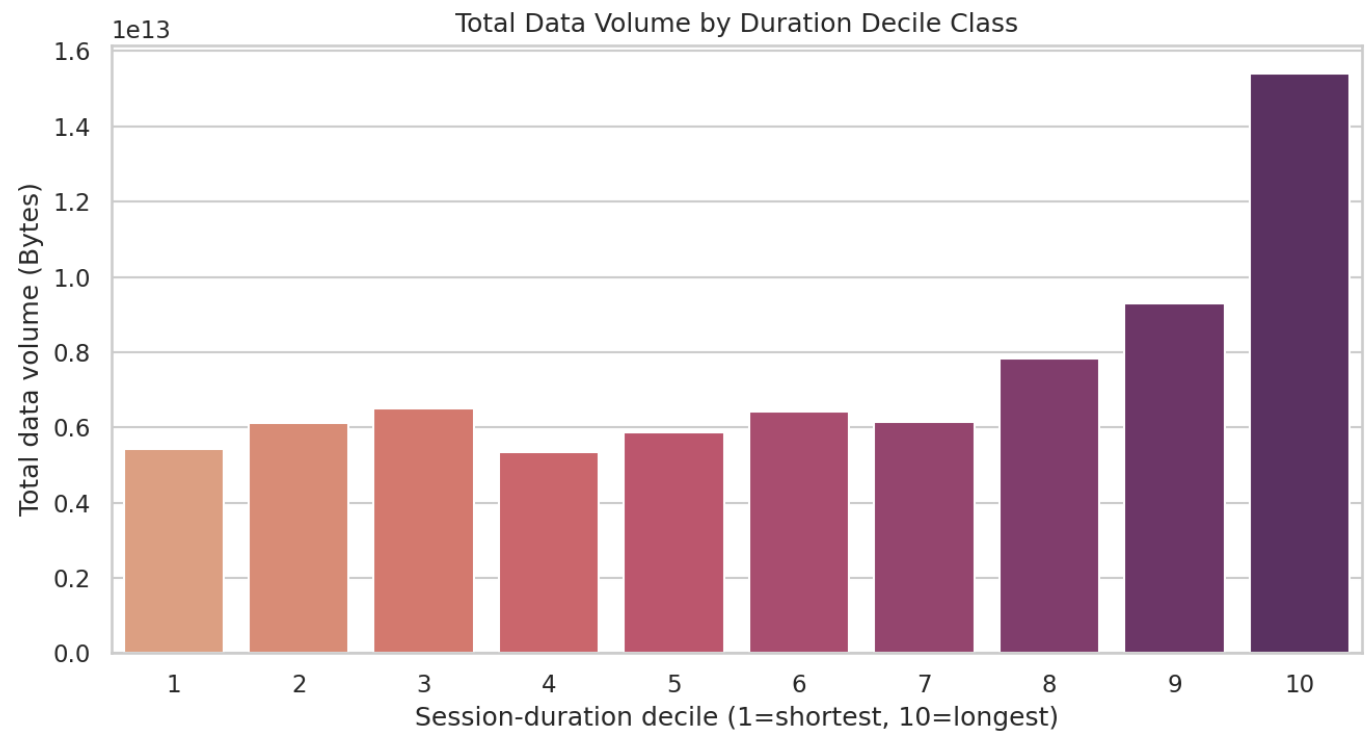
The Huawei B528S-23A -- a fixed-wireless/home-broadband router, not a phone -- is TellCo's single most common device, pointing to a meaningful home-broadband customer segment alongside mobile handsets.

Application Usage: Correlation & PCA



- Application categories are essentially uncorrelated -- usage of one app tells you little about another
- PCA variance is spread evenly (~14% per component) -- no single 'heavy user' axis
- Total data volume is driven almost entirely by Gaming ($r=0.998$)

Data Volume by Session-Duration Decile



Top decile alone accounts for

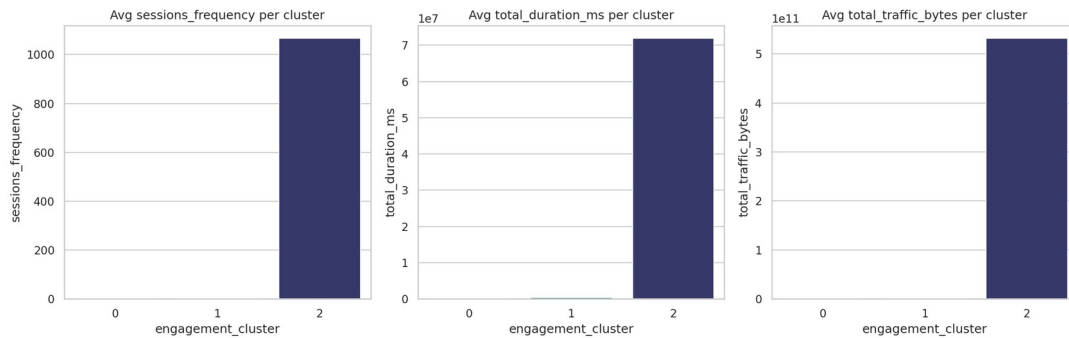
14.0 TB

of total network data volume

Long-session users are disproportionately valuable data consumers -- more than double the 6th decile's volume. These subscribers are prime candidates for premium/unlimited plan upsell.

Engagement Metrics & Top Customers

Engagement was measured per subscriber via three metrics: session frequency, total session duration, and total session traffic (DL+UL bytes).

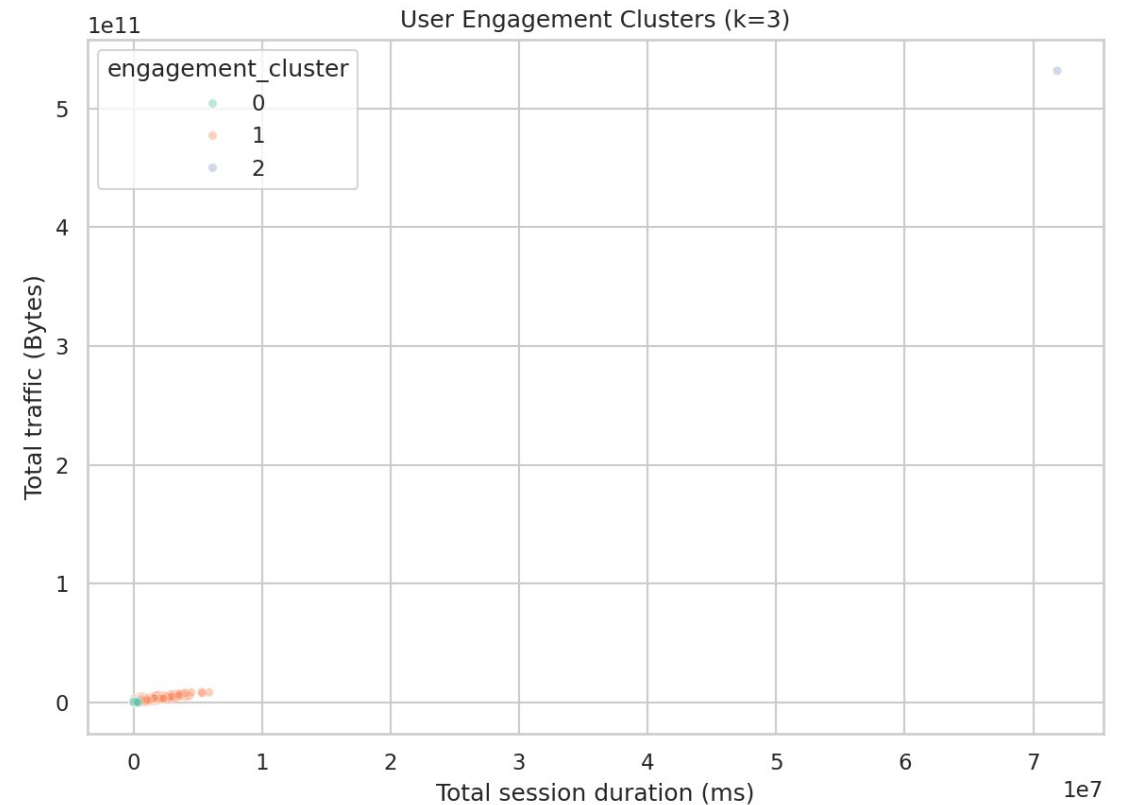
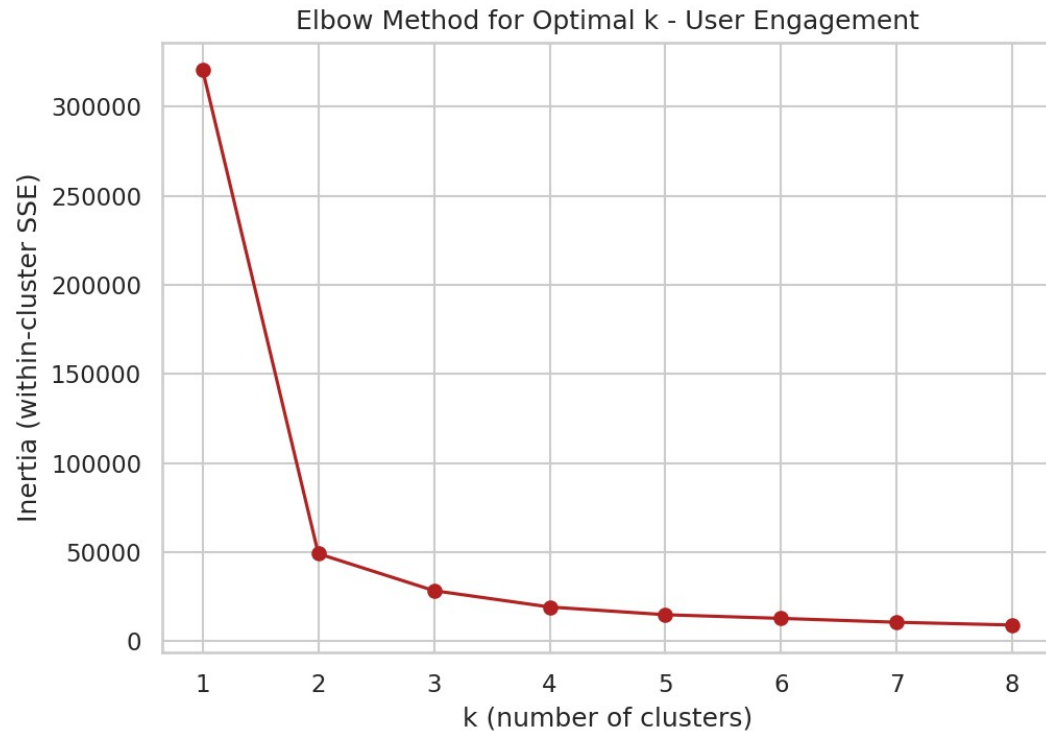


Cluster 2 (the single outlier) dwarfs the scale of clusters 0-1, visually flattening them -- see Slide 10 for the full 3-cluster breakdown.

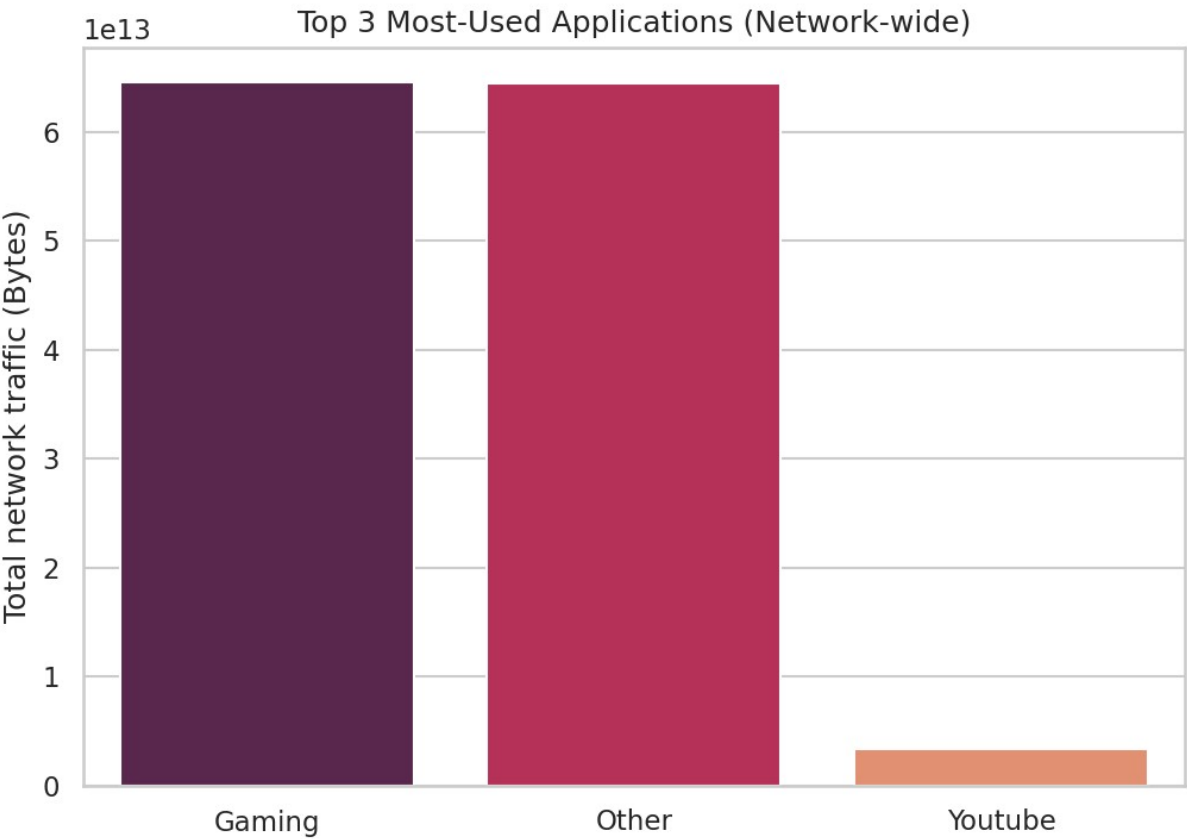
What drives engagement

- One subscriber logged 1,066 sessions in the month vs. a median of ~1 -- an extreme statistical outlier (likely M2M/aggregator line)
- Excluding that outlier, the next-heaviest users show 12-18 sessions -- a natural 'power user' tier
- Gaming and unclassified 'Other' traffic dominate total bytes network-wide

Engagement Segmentation (K-Means, k=3)



Most-Used Applications, Network-Wide



58.7 TB

Gaming total volume

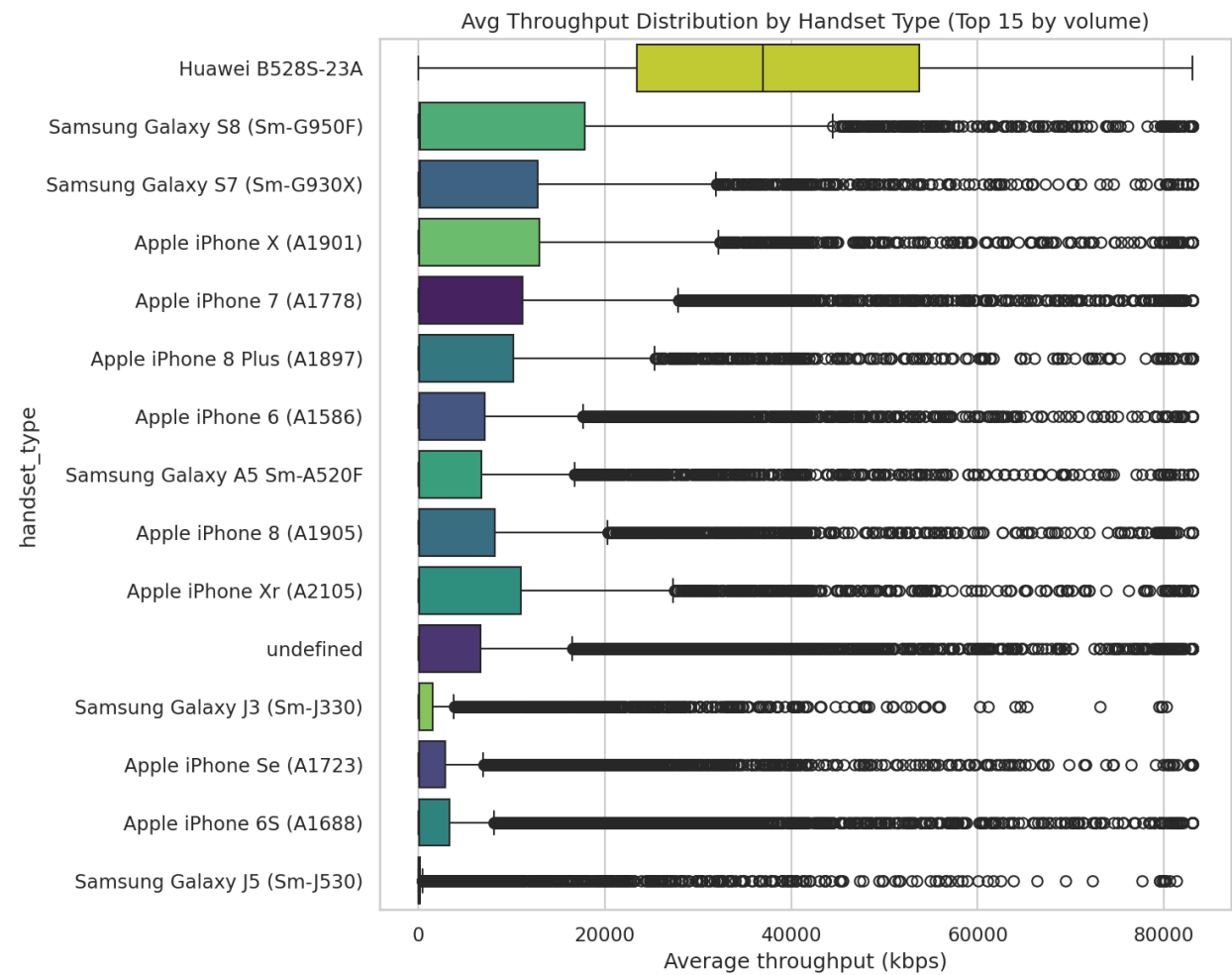
58.6 TB

Other total volume

3.1 TB

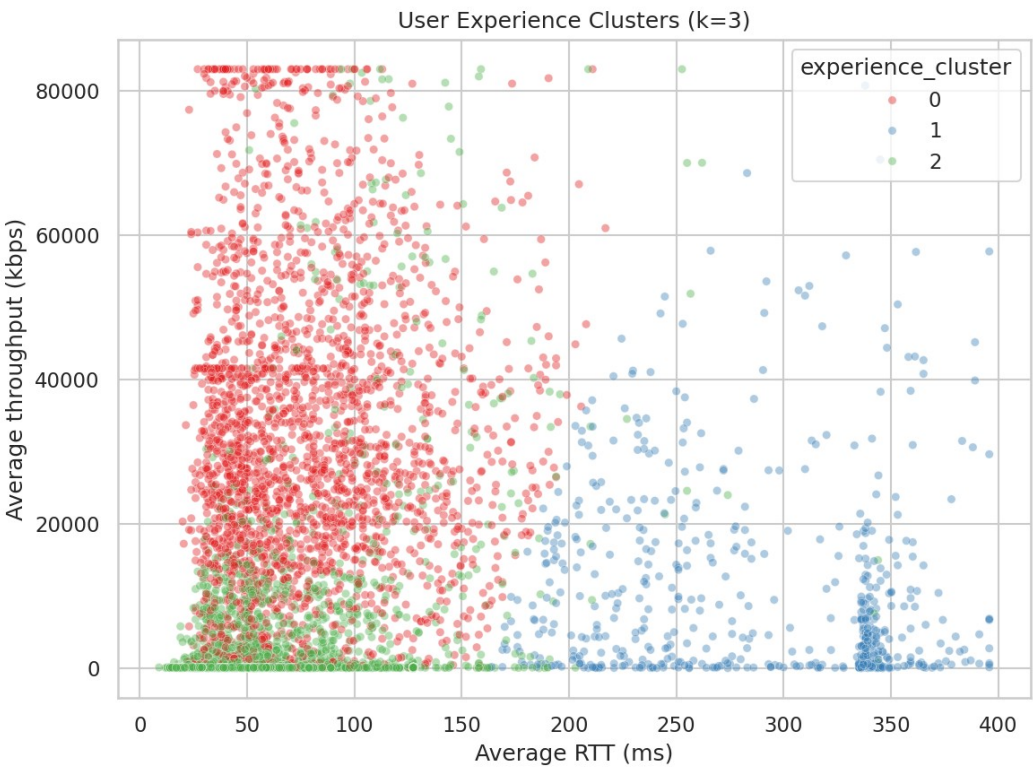
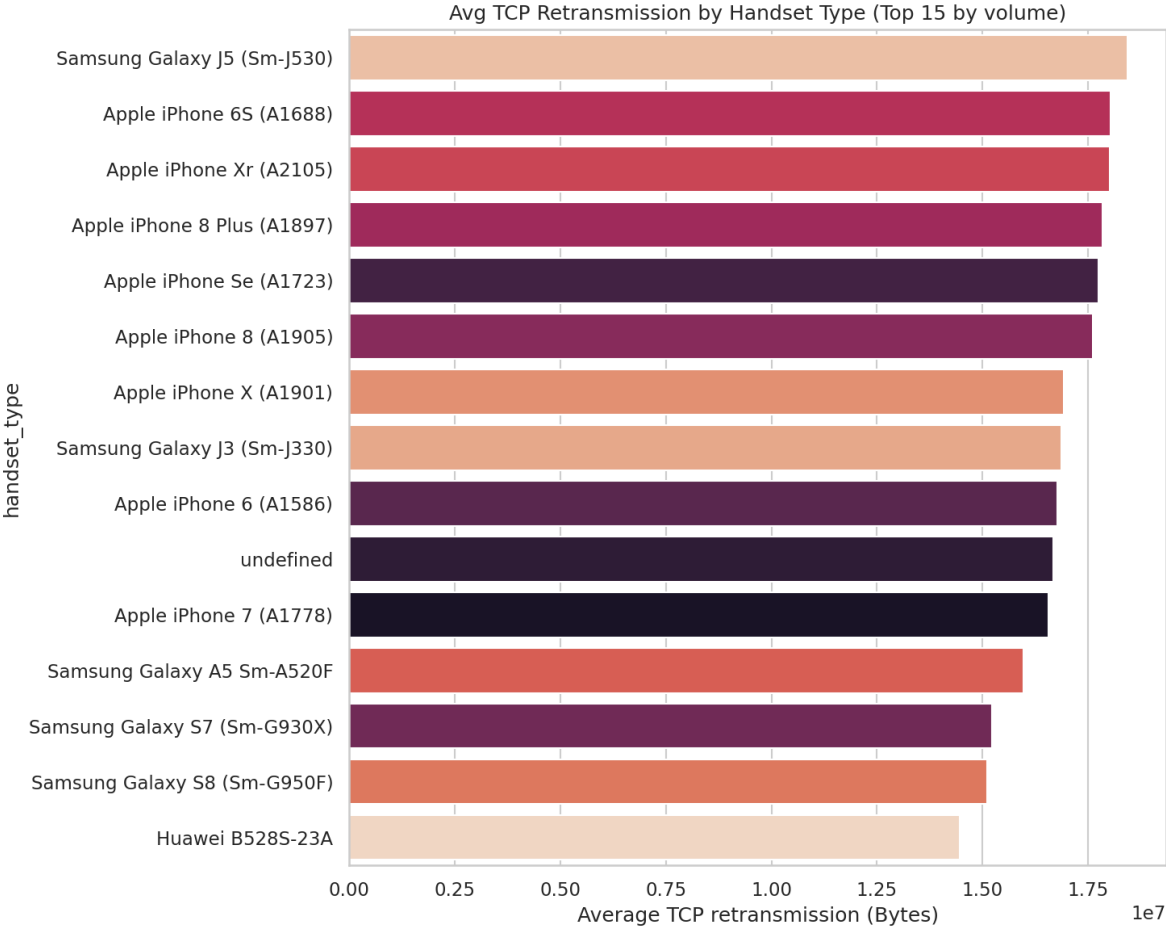
YouTube total volume

Average Throughput by Handset Type



Top 15 handsets by session volume. Spread within a single model suggests real network/coverage variance, not just device capability.

Network Quality: Retransmission & Experience Clusters



Three clean experience clusters emerge: good (low RTT/retrans, high throughput), moderate, and poor. The poor-experience cluster anchors the Task 4 satisfaction scoring.

Scoring Customer Satisfaction

1

Engagement score

Distance from each user to the LEAST-engaged cluster centroid

2

Experience score

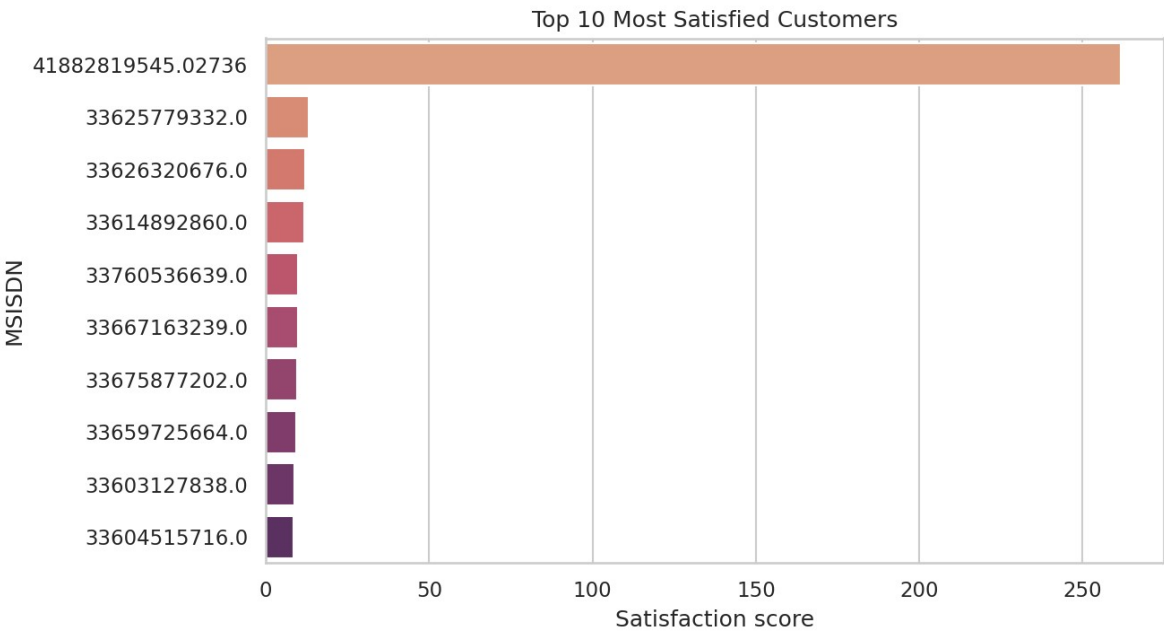
Distance from each user to the WORST-experience cluster centroid

3

Satisfaction score

Average of engagement score and experience score

Top Satisfied Customers & What Drives Satisfaction

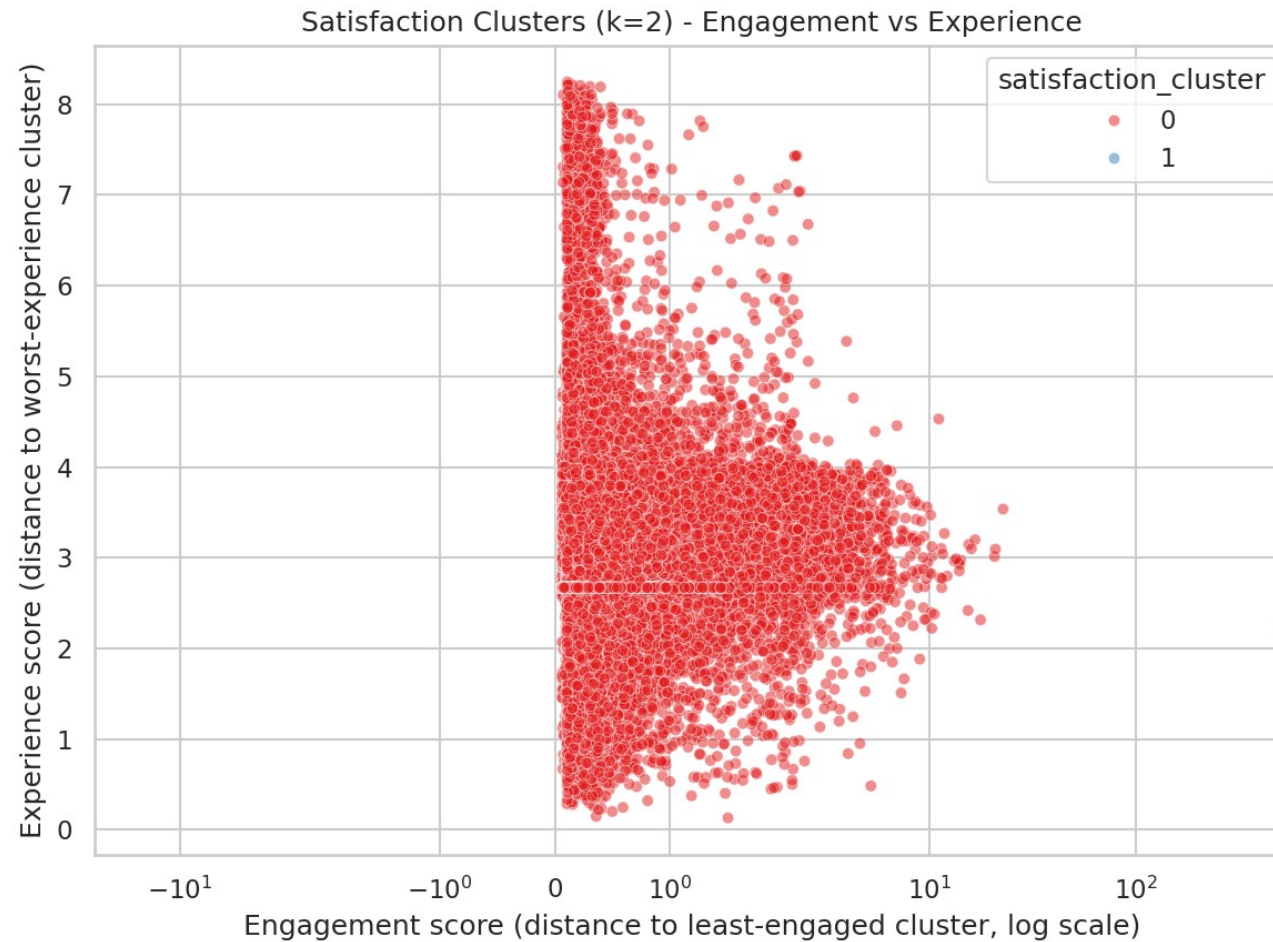


Regression $R^2 = 0.9509$

Feature	Coefficient
sessions_frequency	0.5424
avg_rtt_ms	-0.3039
total_duration_ms	0.246
avg_throughput_kbps	0.1948
avg_tcp_retrans_total_bytes	0.1901
total_traffic_bytes	0.0544

Session frequency and duration are the strongest positive drivers of satisfaction.

Satisfaction Clusters (k=2)



Cluster 0

Avg satisfaction: 1.8376

Avg experience: 3.246

Avg engagement: 0.4292

Cluster 1

Avg satisfaction: 261.7622

Avg experience: 2.3027

Avg engagement: 521.2216

MLOps: Model Deployment & Tracking

1

Versioned runs

Each training run logs code version, start/end time, and parameters

2

Tracked metrics

Train/test R^2 and RMSE recorded per run in `model_tracking_log.csv`

3

Artifacts saved

Coefficients and full run records archived to `reports/model_runs/`

4

Database export

Final scored table (engagement, experience, satisfaction) exported for SQL access

Analysis Limitations

- Single-month snapshot -- trend, seasonality, and churn cannot be assessed
- 55-87% missingness in key network-quality columns (TCP retransmission, HTTP bytes), imputed per brief
- One extreme outlier subscriber dominates engagement/satisfaction clustering and rankings
- No revenue or pricing data provided -- engagement/experience used as value proxies, not profitability
- Custom numpy modeling toolkit used (no scikit-learn/scipy in this environment) -- unit-tested but not cross-validated against scikit-learn
- Database export used SQLite as a stand-in for MySQL (no local MySQL server available)

Buy, With Conditions

- Positive: large, engaged subscriber base with a premium device mix and an underexploited fixed-wireless broadband segment
- Risk: immature network telemetry and measurable per-handset quality variance should be priced into the deal
- Recommended next step: negotiate a network-infrastructure remediation allowance into the purchase price
- Recommended next step: commission a multi-month follow-up analysis to confirm trends before final close

Bottom line: the data supports proceeding with the acquisition, conditioned on a network-quality remediation plan.

References

- TellCo xDR session-level export (source dataset), Nexthikes IT Solutions
- Field Descriptions.xlsx -- field dictionary accompanying the source dataset
- Project code repository: tellco-analysis/ (src/tellco package, scripts/, tests/, dashboard/app.py)
- Full written report: TellCo_Investor_Report.docx
- Interactive dashboard: dashboard/app.py (Streamlit)